CS-2008 Numerical Computing **Quiz 4 - BCS-7A** **Fall 2025**

Name:_____ Roll Number: _____

Marks Distribution: 4 x (1.5) + (5) + (4)

Q1: Why does Richardson's Extrapolation require two different step sizes?

Ans:

Q2: What is the main limitation of the basic Euler method?

Ans:

Q3: How does Improved Euler's Method correct the inaccuracy of the basic Euler method?

Ans:

Q4: Write the formula of second order Runge–Kutta method.

Ans:

Q5:

1. **Richardson's Extrapolation primarily improves accuracy by:**
   - **A. Increasing the order of the derivative**
   - **B. Cancelling the leading term in the error expansion**
   - **C. Using backward differences**
   - **D. Reducing memory usage**

2. **Basic Euler Method uses slope information from:**
   - **A. Midpoint**
   - **B. Both endpoints**
   - **C. Only the beginning of the interval**
   - **D. Tangent at the previous two points**

3. **The local truncation error of 2nd-order RK is:**
   - **A.** $\boldsymbol{O(h)}$
   - **B.** $\boldsymbol{O(h^2)}$
   - **C.** $\boldsymbol{O(h^3)}$
   - **D.** $\boldsymbol{O(h^4)}$

4. **RK4 uses how many evaluations per step?**
   - **A. 2**
   - **B. 3**
   - **C. 4**
   - **D. 5**

5. **Which method is known as a second-order RK method?**
   - **A. Euler**
   - **B. Improved Euler**
   - **C. RK3**
   - **D. RK4**

Q6 True / False

1. **Richardson's Extrapolation can only be used for derivatives.**
2. **Euler and RK methods are both one-step methods.**
3. **Higher-order RK methods always require calculating higher-order derivatives.**
4. **Improved Euler method is more accurate because it uses an average slope.**